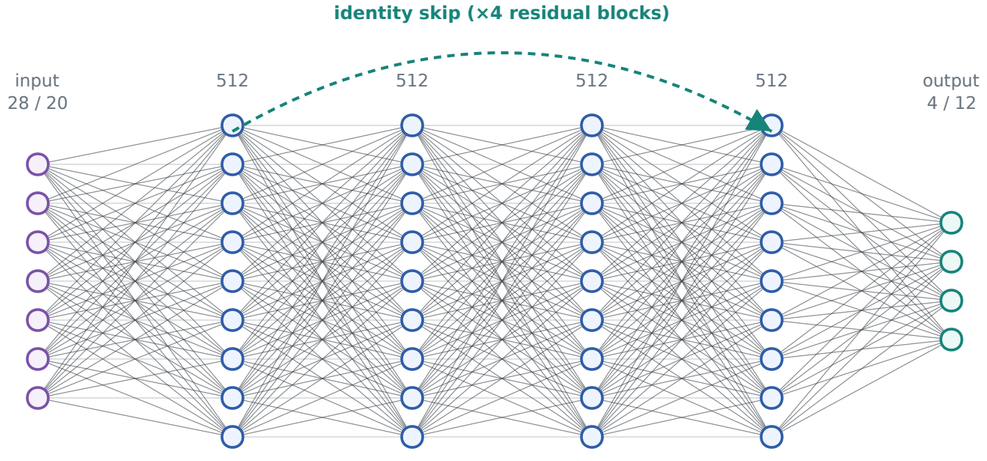
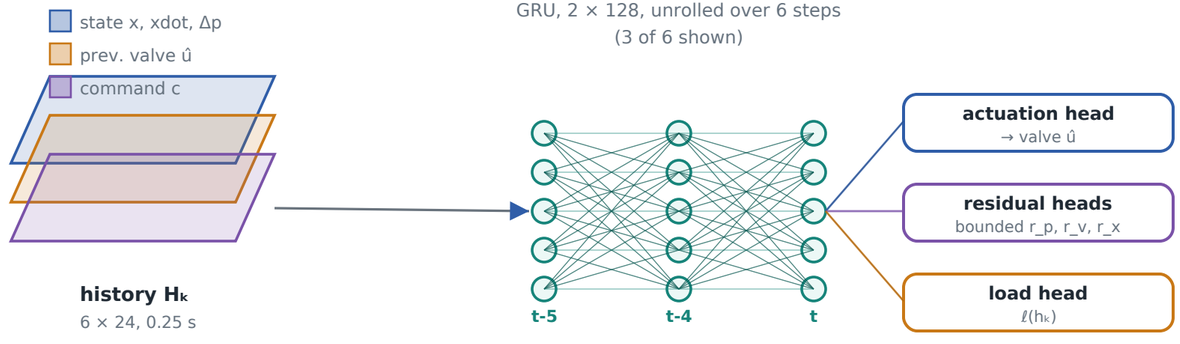
(a) ID / FD residual network (neuron view)



SiLU + LayerNorm + Dropout 0.02 between linear layers

shared trunk: ID predicts Δu (4-D), FD predicts [Δx, Δxdot, ΔΔp] (12-D)

(b) Command-conditioned ROM encoder (neuron view)



gray fan lines = full connectivity between adjacent layers; colored slabs = distinct feature groups in the 24-D input, in the style of the multi-feature input stacks used by Liu et al. (2025) for hydraulic-manipulator dynamics networks.